MACH1 — Battery Lifecycle Model

```
SoH(t, EFC) = 1.02 - A_cal·t^α·Arrhenius(T)·k_SOC - k_cyc·DoD^β·Arrhenius(T)·(EFC·8)
```

Everything below is a decomposition of that one line: a fitted offset, then two degradation terms — calendar first, cycle second — each scaled by its own `Arrhenius(T)`.

Calibration basis — the cell the model is fitted to, and the defaults it ships with

- **Cell** · Sony/Murata US26650FTC1 — 3 Ah LFP/graphite cylindrical
- **Basis** · 885 days · 17 test points × 3 cells · <2.2% max error on capacity fade
- **Setup** · α = 0.50 · β = +1.00 · DoD ≈ 57%
- **Source** · Naumann et al. 2018, J. Energy Storage 17, 153–169

Table 1 — Where every term in the equation comes from

Read left to right, in the equation's own order: offset, then calendar term, then cycle term. Seven factors are fitted or cited and locked; two are contested in the open literature, and both are what make the next table a range.

Symbol	Term	Value	Source	Status
`1.02`	cal · offset	1.02	fitted intercept	locked
`- A_cal`	cal · coefficient	—	Naumann 2018	locked
`· t^α`	cal · time exponent	**0.42 → 1.00**	Lewerenz '17 · Naumann '18 · Gasper '21 · Liu '18	**⚠ CONTESTED — 32 kWh pack**
`· Arrhenius(T)`	cal · temperature	Ea 17.126 kJ/mol (capacity)	Naumann 2018	locked
`· k_SOC`	cal · SOC factor	0.6140 @ 66% mean SoC	Naumann 2018 (literal form)	locked
`- k_cyc`	cyc · coefficient	—	Naumann 2018	locked
`· DoD^β`	cyc · DoD exponent	**−0.40 → +1.00**	Gauthier '22 · Eldesoky '22 · Eldesoky '20	**⚠ CONTESTED — 7 kWh pack**
`· Arrhenius(T)`	cyc · temperature	Ea 71.827 kJ/mol (resistance) — **4.2× the capacity Ea**	Naumann 2018	locked
`· EFC · 8`	duty · full-charge count	~200 EFC/yr → 1548 @ yr 8	duty: 12.84 kWh/100 km · 25.68 kWh/day	locked

Slide caption — one equation, nine factors: seven fitted or cited and locked, two still argued over in print.

Closing the contested rows — the validation programme that resolves them, not splits them

- **Test** · 6 cells · 40–45 °C · 90–100% SoC · ≥18 months

> ***Caption text — Table 1***
> *The two contested rows are the entire uncertainty budget. On **α**, Lewerenz '17 measured linear fade across 50 LFP cells and got 1.00, while Naumann '18 fitted √t and got 0.50, with Gasper '21 at ~0.45 and Liu '18 at 0.42 — a **2.3× spread that swings the pack from 44 to 76 kWh**. On β**, Gauthier '22 reports +1.0 at 40 °C, Eldesoky '22 finds no DoD dependence at all below a 4.06 V cut-off, and Eldesoky '20 reports −0.40 at 55 °C, which swings the pack by 7 kWh. So the bands in Table 2 are not vagueness in our model — every band traces to exactly one unresolved literature disagreement, and α is the dominant one; the validation programme above is built to close both rather than split the difference.*

Table 2 — What the equation predicts

Pack size required to meet fleet duty, and degradation at year 8, by cell temperature

Cell temp	Pack required (band)	Pack @ defaults	SoH @ yr 8 (band)	SoH @ defaults	R_inc @ yr 8	Verdict
26 °C	37.1–53.3 kWh	37.2 kWh	59.5–85.5%	85.2%	+14.0%	✓ Wide margin
40 °C	40.6–74.6 kWh	41.0 kWh	42.5–78.1%	77.3%	+50.9%	✓ Achievable
48 °C ★	43.3–100.1 kWh	44.2 kWh	31.7–73.2%	71.8%	+101.3%	⚠ Band too wide
55 °C	46.2–**140**+ kWh	47.8 kWh	22.0–68.6%	66.4%	+179.8%	✗ Infeasible

★ 48 °C is our design point (liquid-plate cooling). At default parameters our **45 kWh** pack meets duty, with 193 EFC/yr and 1548 EFC by year 8.
The ✗ at 55 °C refers to the band's worst corner (β = −0.40 with α = 1.00), not to our defaults — the default there still closes at **47.8 kWh**. Every temperature in this table has a finite default answer; what runs away with heat is the *width* of the band. At 55 °C the worst corner passes the model's 140 kWh search ceiling — roughly 3× our 45 kWh sedan — and we did not raise that ceiling to manufacture a finite answer.

Slide caption — our default closes at every temperature; it is the width of the band, not the answer, that runs away with heat.

> ***Caption text — Table 2***
> *Every band here is Table 1's two contested parameters carried forward — α contributes 32 kWh and β contributes 7 kWh — so the width is the literature disagreement, not vagueness in our model; the defaults are finite and specific at every temperature, which is the whole reason we default rather than average. What degrades with temperature is the worst corner of the band, and at 55 °C that corner passes the model's 140 kWh search ceiling, roughly 3× our 45 kWh sedan. We did not raise the ceiling to manufacture a finite answer, because a corner where the disagreement stops being resolvable is precisely the thing we are designing away from. That is why the vehicle is specified to a 48 °C cell on liquid-plate cooling: far enough from that corner that the disagreement still resolves into a pack we can actually build.*

Warranty claim boundary — two separate claims, kept separate on purpose

Time-to-80% at 40 °C differs **3.6×** between Naumann (13.3 yr) and Lewerenz (3.7 yr), so **8-year / 80% is not a defensible claim** on this evidence. First life (~15 yr at our duty) is solid; it is the **warranty** that is the uncertain claim. We size and cool the pack against first-life physics, and we do not sell an 8-year warranty the literature will not carry.